

16.4: SambaNova SN40L: A 5nm 2.5D Dataflow Accelerator with Three Memory Tiers for Trillion Parameter AI

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Abstract:
The SN40L is the latest-generation Reconfigurable Dataflow Unit (RDU) from SambaNova Systems built for modern AI training and inference applications [1], [2]. The SN40L c... [Show More](#)

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The SN40L is the latest-generation Reconfigurable Dataflow Unit (RDU) from SambaNova Systems built for modern AI training and inference applications [1], [2]. The SN40L combines on-chip dataflow flexibility with a three-tier memory system: on-chip SRAM, on-package High Bandwidth Memory (HBM), and off-package DRAM (Fig. 16.4.7). The SN40L employs 3x CoWoS-S interposer technology featuring two identical accelerator dies and HBM memory modules. To enable both high-bandwidth and high-capacity memory accesses from a single chip, along with peer-to-peer fabric and die-to-die connectivities, a dual-die approach is adopted, where the size of each die is 600mm² (Fig. 16.4.1). Furthermore, a sub-reticle size dual-die design optimizes performance, thermal efficiency, and cost. The SN40L accelerator dies are fabricated using TSMC's 5nm process. Each socket contains a total of 102 billion transistors, delivering 640 BF16 TFLOPs and 520MB on-chip SRAMs using high-density memory cells.

Each SN40L die consists of four types of programmable units arranged in a 2D mesh interconnect: 520 Pattern Compute Units (PCUs), 520 Pattern Memory Units (PMUs), Switches, and Address Generation and Coalescing Units (AGCUs) (Fig. 16.4.2). PCUs consist of a pipeline of vectorized ALUs that deliver all the compute teraflops (Fig. 16.4.3). ALUs in the PCU can be programmed to operate as either a systolic array for matrix multiplications, or as a static dataflow pipeline for streaming operations. Each PCU also has a tail ALU stage that natively supports complex operations like exponent, sigmoid, and tanh. PMUs each contain 512KB of programmer-managed on-chip SRAM with multiple banks to allow concurrent reads and writes with arbitrary access patterns at high throughput (Fig. 16.4.3). PMUs also contain specialized ALUs to accelerate address computation, and data alignment units to accelerate operations like transpose. Switches support packet-switched scalar and vector data interconnects, and a statically programmed bit-level control interconnect. Packet-switched interconnects support point-to-point and multicast paths. Packets are routed with hardware dimension-ordered routing (DOR), with a provision for software override. AGCUs enable high throughput off-chip address generation as well as virtualization for off-chip memory management. A graph control interface enables RDUs to load and control graph orchestration without host involvement. A hardware-managed Top-Level Network (TLN) connects AGCUs to the 10 subsystem components like HBM, and PCIe links to other RDUs and the host CPU. Data can be directly streamed from one RDU to another RDU using a token-based Peer-to-Peer (P2P) protocol.

Compute graphs are executed on RDUs in a pipelined manner where compute is overlapped with communication by default. Subtensors, or tiles, of input Tensors are loaded by AGCUs from HBM or DDR into one or more PMUs ([Fig. 16.4.2](#)). Data arrives in packets into PMUs, where ALUs are programmed with specific read and write access patterns to access data in the SRAMs. Once a data tile is written into a PMU, a control token starts the read operation from the PMU. Data from the PMU is routed to its on-chip destination units - PCUs, PMUs, or AGCUs - via switches along either dimension-ordered or software-determined paths ([Fig. 16.4.3](#)). Each unit is programmed with input data and control dependencies. Units begin processing its inputs and producing outputs when its dependencies are met. For instance, a PCU programmed to add two Tensors begins execution when it has received at least one data packet for each input. Loading data tiles from AGCUs, reads and writes in PMUs, and compute in PCUs are all fully decoupled and overlapped with each other.

DDR and HBM memories are exposed to software directly as two separate memory spaces. Software allocates Tensors requiring high bandwidth access - like weights and KV caches in generative inference - in HBM. DDR is used to store Tensors requiring high capacity and low bandwidth access. For instance, training applications can use DDR to store activation checkpoints generated during the forward pass that is required during backpropagation. In generative inference and agentic AI applications, DDR can be used to store parameters of hundreds of models along with persistent KV cache values [\[5\]](#). The most frequently used models are then moved to HBM dynamically by software. With advances in agentic AI and test time inference scaling [\[3\]](#), [\[4\]](#), modern AI applications perform inference on several models to collectively accomplish a complex task. With DDR and HBM, a single SN40L node with 8 sockets (SN40L-8) lowers both model hosting and switching costs [\[2\]](#), and is able to host and serve trillions of parameters efficiently in a cost-effective manner, enabling much higher Performance /Total Cost of Ownership (TCO) ([Fig. 16.4.5](#)).

The increasing demand for computational capability in LLMs has posed a significant challenge in meeting both thermal and electrical power demands in many conventional data centers. While both hardware and software power capping solutions can be employed to limit the power at the cost of reduced performance, the fast-moving power variation behavior of LLM workloads makes it difficult to optimize performance for every type of workload. This often results in static changes to lower-performing states to stay within the constraint. For example, a typical inference workload consists of two main phases: the prompt phase, which is a compute intensive phase that processes a large input matrix, requiring significant computational resources, and the generation phase, which sequentially generates tokens using relatively lighter and more stable computation, and consumes less power. The compute interval of the prompt phase can span from milliseconds to seconds, depending on the input context length and batch size. Therefore, the system needs to be dynamically adaptive to control peak power consumption to stay within the electrical power limit of a rack in the data center during the prompt phase while running at full frequency during the generation phase.

To control both peak and average power under dynamically changing workloads in inference or training models, the SN40L has implemented an on-chip Power Estimation Unit (PEU). The PEU is capable of estimating power consumption in the sub-microsecond range and automatically adjusting operating conditions to achieve maximum performance on any workload within electrical power limits, all without software assistance. The PEU monitors the activities of all tile components, totaling approximately 3000 PCUs, PMUs, and switches, and aggregates the power consumption of each component. Within each unit, a power calculator unit tracks opcode, stage activity, data toggle rate, and zero data detection to ensure accurate power estimation. Additionally, for PMU and switch units, it monitors SRAM operation and network port activity to estimate power. The power consumption of each unit is then collected in a hierarchical daisy chain manner to minimize hardware costs. Each unit contains weights and biases for power calculation, enabling post-Silicon adjustments. The aggregated power is then converted with effective frequency and voltage during its calculation to obtain actual power values. The power management controller receives these power values to calculate the appropriate operating condition and initiates fast-reacting clock skipping, followed by voltage and frequency scaling. As the PEU-controlled power capping operates within microsecond ranges, it ensures the chip never exceeds electrical power budget, staying within the rack limit. [Figure 16.4.4](#) illustrates the fast transitional peak power staying close to average power (TOP) when the power is capped by 50%. The performance improvement achieved with PEU-based power capping shows up to 39% better token generation performance in inference workloads as it throttles only when power exceeds the capping limit, compared to statically setting the operating frequency and voltage to stay within the power limit. Notably, even compute-intensive training workloads show 11% performance improvement, as fast power management with PEU operates at higher frequencies during communication-heavy or compute-light phases.

The SN40L's dataflow architecture and 3-tier memory system are designed to achieve energy efficiency in computing large language models (LLMs) compared to GPUs [6]. The streaming dataflow architecture enables two key optimizations. First, it allows massive kernel fusion, integrating the entire decoder layer into a single kernel call [1], [2]. Second, it reduces kernel launch overhead by converting repeated invocations of this fused kernel into a single call. This minimizes off-chip memory accesses, resulting in power savings on both the accelerator and HBM dies. Additionally, the dataflow architecture minimizes the power overhead associated with core-based ISA accelerators. These features enable SN40L to deliver about 3x better energy efficiency compared to the Nvidia H100 GPU across Llama3.1 8B, 70B, and ViT model inference (Fig.16.4.5). Dataflow also enables better utilization of on-chip compute teraflops during training, achieving a 2x speedup over H100 in training vision transformers (ViT) (Fig.16.4.5). A typical 8-socket SN40L system can host models with up to 5 trillion parameter models, compared to 0.27 trillion parameter models on GPUs with the same socket count [7], all within millisecond model switching latencies. This unique capability of direct DDR access from the SN40L provides power, space, and cost benefits for running a large number of expert models by breaking the memory wall of GPUs. Finally, reduced power density allows the SN40L to maximize its compute resources in air-cooled systems (Fig.16.4.6). This eliminates the need for costly and complex cooling solutions and enables deploying SN40L in datacenters worldwide.



Figure 16.4.1:
SN40L socket components and memory tiers.



Figure 16.4.2:
SN40L die with PCUs, PMUs, switches, and AGCUs in a 2D mesh interconnect.



Figure 16.4.3:
PCU (top) and PMU (bottom) architecture in SN4DL.



Figure 16.4.4:
Unconstrained vs. constrained power profiling during Llama278 training.



Figure 16.4.5:
SN40L performance and energy vs. DGX H100.



Figure 16.4.6:
Temperature and utilization characteristics of GPT-138 training on SN40L.



Figure 16.4.7:
SN40L die and board photo showing chip and three tiers of memory.

